

Symbol		Unit	Meaning
<i>Indices</i>			
i, j		Token	Input (sold) and output (bought) endpoint tokens; the ordered pair is $i \rightarrow j$.
k		Token	Candidate vehicle token used as the route intermediate; $k \notin \{i, j\}$.
ℓ, p		Token / pool	ℓ indexes tokens in cross-token sums; p indexes liquidity pools.
t, w		UTC day / UTC week	t indexes calendar days; w indexes calendar weeks in settlement variables.
q		USD	Input quote notional. Un-suffixed route-cost columns use $q = \$10,000$.
r		Route unit	Receipt-audited coherent multihop route unit indexed by r .
<i>Route and liquidity aggregates</i>			
A_t, DVol_t		USD	A_t is total realized USD volume across all direct and indirect route units on day t . DVol_t restricts it to direct routes, so $0 \leq \text{DVol}_t \leq A_t$.
$\text{IVol}_t, \text{IVol}_{k,t}$		USD	IVol_t restricts A_t to indirect routes, so $0 \leq \text{IVol}_t \leq A_t$. $\text{IVol}_{k,t}$ further restricts that volume to routes using vehicle k , so $0 \leq \text{IVol}_{k,t} \leq \text{IVol}_t$.
$N_t^{\text{route}}, N_{k,t}^{\text{src}}, N_{k,t}^{\text{sink}}$		Route count	N_t^{route} counts all day- t intent routes. $N_{k,t}^{\text{src}}$ and $N_{k,t}^{\text{sink}}$ restrict that count to source and sink roles for k ; each lies between 0 and N_t^{route} .
$N_t^B, N_{k,t}^B$		Route-unit count	N_t^B restricts N_t^{route} to indirect routes. $N_{k,t}^B$ further restricts that count to routes using k , so $0 \leq N_{k,t}^B \leq N_t^B \leq N_t^{\text{route}}$; superscript B denotes bridged (indirect) routes.
$\mathcal{A}_t, \mathcal{A}_t^k, \mathcal{M}_t^k$		Sets of token pairs	$\mathcal{A}_t$ is the set of endpoint pairs active in indirect routes on day t . $\mathcal{A}_t^k$ restricts it to pairs using k , and $\mathcal{M}_t^k$ further restricts it to pairs for which k has the largest candidate-vehicle volume; $\mathcal{M}_t^k \subseteq \mathcal{A}_t^k \subseteq \mathcal{A}_t$.
$\text{Vol}_{k,t}^{\text{in}}, \text{Vol}_{k,t}^{\text{out}}$		USD	$\text{Vol}_{k,t}^{\text{in}}$ and $\text{Vol}_{k,t}^{\text{out}}$ are inbound and outbound route-leg volumes assigned to token k on day t .
$\text{Vol}_{k,t}^{\text{src}}, \text{Vol}_{k,t}^{\text{sink}}$		USD	Source- and sink-role restrictions of A_t for token k : $0 \leq \text{Vol}_{k,t}^{\text{src}} \leq A_t$ and $0 \leq \text{Vol}_{k,t}^{\text{sink}} \leq A_t$.
$\mathcal{K}, L_{k,t}$		Set of tokens / USD	$\mathcal{K}$ is the candidate-vehicle set. $L_{k,t}$ is liquidity linked to candidate $k \in \mathcal{K}$ on day t .
$\mathcal{L}_{k,t}, \text{TVL}_{p,t}$		Set of pools / USD	$\mathcal{L}_{k,t}$ is the set of eligible pools linked to k on day t ; $\text{TVL}_{p,t}$ is total value locked in pool p on that day.
R_t^{WETH}		Daily log-return fraction	Day- t log return of the WETH price used to construct downside stress.
<i>Route-cost quote objects</i>			
$\mathcal{P}_{k,t,q}$		Set of token pairs	Ordered endpoint pairs eligible for k and quoted on day t at notional q .
$\mathcal{D}_{k,t,q}, \mathcal{V}_{k,t,q}, \mathcal{C}_{k,t,q}, \mathcal{T}_{k,t,q}, \mathcal{W}_{k,t,q}$		Sets of token pairs	$\mathcal{D}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ and $\mathcal{V}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ restrict the broad pair set to pairs with a direct route or a route via k . $\mathcal{C}_{k,t,q} = \mathcal{D}_{k,t,q} \cap \mathcal{V}_{k,t,q}$, $\mathcal{T}_{k,t,q} \subseteq \mathcal{D}_{k,t,q}$ is thin-direct support, and $\mathcal{W}_{k,t,q} \subseteq \mathcal{C}_{k,t,q}$ has a vehicle-route cost advantage.
$D_{i,j,q,t}, V_{i,j,k,q,t}, T_{i,j,q,t}$		Indicator (0/1)	Indicators for direct-route availability (D), route-through- k availability (V), and an executable direct route returning less than $0.9q$ (T).
$O_{i,j,q,t}^D, O_{i,j,k,q,t}^V$		USD	Quoted output values; superscripts D and V denote direct and vehicle routes.
$\Delta C_{i,j,k,q,t}$		Basis points	10,000 $(O_{i,j,k,q,t}^V - O_{i,j,q,t}^D) / O_{i,j,q,t}^D$ on $\mathcal{C}_{k,t,q}$.
<i>Settlement objects and operators</i>			
$\mathcal{R}_{k,w}$		Set of route units	Receipt-audited matched route units using vehicle k in UTC week w .
$\mathcal{R}_{k,w}^{\text{transfer}}$		Set of route units	Members whose receipt logs a transfer of vehicle k ; $\mathcal{R}_{k,w}^{\text{transfer}} \subseteq \mathcal{R}_{k,w}$.
$ \mathcal{S} , \mathbf{1}_{\{\cdot\}}$		Count / indicator (0/1)	Cardinality of any finite set $\mathcal{S}$ and an indicator equal to one when its subscripted condition is true.
Δ_7		7 days	Seven-day forward change in any daily variable X_t : $\Delta_7 X_t = X_{t+7} - X_t$.

Variable	Formula	Unit	Data column	Definition
<i>Vehicle-use measures</i>				
VehicleShare $_{k,t}$	$\frac{\text{IVol}_{k,t}}{\text{IVol}_t}$	Fraction (0–1)	bridge_share	Fraction of day- t indirect-route USD volume routed through k .
AllRouteVehicleShare $_{k,t}$	$\frac{\text{IVol}_{k,t}}{A_t}$	Fraction (0–1)	all_route_bridge_share	Fraction of day- t all-route USD volume routed indirectly through k .
VehicleCountShare $_{k,t}$	$\frac{N_{k,t}^B}{N_t^B}$	Fraction (0–1)	bridge_count_share	Fraction of day- t indirect route units that use k .
PairCoverage $_{k,t}$	$\frac{ \mathcal{A}_t^k }{ \mathcal{A}_t }$	Fraction (0–1)	pair_coverage	Fraction of active indirect-route endpoint pairs that use k .
MainVehiclePairShare $_{k,t}$	$\frac{ \mathcal{M}_t^k }{ \mathcal{A}_t }$	Fraction (0–1)	pair_main_vehicle_share	Fraction of active indirect-route endpoint pairs for which k has the largest candidate-vehicle volume.
IVol $_{k,t}$		USD	bridge_volume_usd	Sum of realized USD volume over indirect routes using vehicle k on day t .
<i>Network and route-denominator controls</i>				
VolShare $_{k,t}$	$\frac{\text{Vol}_{k,t}^{\text{in}} + \text{Vol}_{k,t}^{\text{out}}}{\sum_{\ell} (\text{Vol}_{\ell,t}^{\text{in}} + \text{Vol}_{\ell,t}^{\text{out}})}$	Fraction (0–1)	vol_share	$\text{Vol}_{k,t}^{\text{in}}$ and $\text{Vol}_{k,t}^{\text{out}}$ are inbound and outbound route-leg USD volumes for k ; ℓ indexes every token in the day- t network.
Betweenness $_{k,t}$	$\frac{N_{k,t}^B}{N_t^{\text{route}} - N_{k,t}^{\text{src}} - N_{k,t}^{\text{sink}}}$	Fraction (0–1)	betweenness_centrality	Fraction of day- t intent routes on which k is intermediate, excluding routes where k is the source or sink. Superscripts identify each route role.
Betweenness $_{k,t}^{\text{vol}}$	$\frac{\text{IVol}_{k,t}}{A_t - \text{Vol}_{k,t}^{\text{src}} - \text{Vol}_{k,t}^{\text{sink}}}$	Fraction (0–1)	volume_weighted_betweenness	USD-volume analogue of Betweenness $_{k,t}$ using the same route universe and source/sink exclusions.
A_t	$\text{DVol}_t + \text{IVol}_t$	USD	daily_all_route_volume_usd	Total realized USD volume across direct and indirect route units on day t .
IVol $_t$		USD	daily_indirect_route_volume_usd	Total realized USD volume across indirect route units on day t .
IndirectRouteShare $_t$	$\frac{\text{IVol}_t}{A_t}$	Fraction (0–1)	indirect_route_share	Fraction of day- t all-route USD volume executed through indirect routes.
<i>Liquidity measures</i>				
$L_{k,t}$	$\sum_{p \in \mathcal{L}_{k,t}} \text{TVL}_{p,t}$	USD	vehicle_linked_liquidity_usd	Total eligible Uniswap V3 pool TVL linked to k after filtering subgraph TVL outliers.
LPConc $_{k,t}$	$\frac{L_{k,t}}{\sum_{\ell \in \mathcal{K}} L_{\ell,t}}$	Fraction (0–1)	lp_concentration	Candidate k ’s share of total vehicle-linked liquidity on day t .
LogVehicleLiquidity $_{k,t}$	$\ln(1 + L_{k,t})$	Natural-log points	log_vehicle_linked_liquidity	Natural log of one plus $L_{k,t}$ measured in USD.
<i>Route-cost opportunity measures</i>				
DirectAvailable $_{k,t,q}$	$\frac{ \mathcal{D}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	direct_available_share	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $D_{i,j,q,t} = 1$. The un-suffixed data column fixes $q = \$10,000$.
VehicleAvailable $_{k,t,q}$	$\frac{ \mathcal{V}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	vehicle_available_share	Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $V_{i,j,k,q,t} = 1$, so both legs through k are executable.
NoDirectVehicleAvailable $_{k,t,q}$	$\frac{ \mathcal{V}_{k,t,q} \setminus \mathcal{D}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	no_direct_vehicle_available_share	Fraction of pairs with $D_{i,j,q,t} = 0$ and $V_{i,j,k,q,t} = 1$: no executable direct route but an executable route through k .
DirectDepth $_{k,t,q}$	$\text{median}_{\mathcal{D}_{k,t,q}} \left(O_{i,j,q,t}^D / q \right)$	Output USD per input USD	direct_depth_median	Median direct-route output ratio $O_{i,j,q,t}^D / q$ over executable-direct pairs $(i, j) \in \mathcal{D}_{k,t,q}$.
RouteCostAdvantage $_{k,t,q}$	$\text{median}_{\mathcal{C}_{k,t,q}} \Delta C_{i,j,k,q,t}$	Basis points	route_cost_advantage_median_bps	Median $\Delta C_{i,j,k,q,t}$ over common-support pairs $(i, j) \in \mathcal{C}_{k,t,q}$; positive values favor the route through k .
VehicleBeatsDirect $_{k,t,q}$	$\frac{ \mathcal{W}_{k,t,q} }{ \mathcal{C}_{k,t,q} }$	Fraction (0–1)	vehicle_beats_direct_share	Fraction of $(i, j) \in \mathcal{C}_{k,t,q}$ for which $\Delta C_{i,j,k,q,t} > 0$, equivalently $O_{i,j,k,q,t}^V > O_{i,j,q,t}^D$.
ThinDirectShare $_{k,t,q}$	$\frac{ \mathcal{T}_{k,t,q} }{ \mathcal{P}_{k,t,q} }$	Fraction (0–1)	thin_direct_share	Fraction of pairs with $D_{i,j,q,t} = 1$ and $O_{i,j,q,t}^D / q < 0.9$.
<i>Stress and dynamic variables</i>				
Stress $_t$	$\max\{-R_t^{\text{WETH}}, 0\}$	Daily log-return fraction	stress_downside	Positive part of the negative day- t WETH log return.
StressEvent $_t$	$\mathbf{1}_{\{\text{Stress}_t \geq 0.08\}}$	Indicator (0/1)	stress_event_8pct	Equals one when Stress $_t$ is at least 0.08 on day t .
VehicleShare $_{k,t+7}$		Fraction (0–1)	future_bridge_share_t7	Vehicle share for token k seven calendar days after day t .
$\Delta_7 \text{VehicleShare}_{k,t}$	$\text{VehicleShare}_{k,t+7} - \text{VehicleShare}_{k,t}$	Fraction points	delta_bridge_share_t7	Seven-day forward change for vehicle k , measured from day t .
<i>V4 settlement implementation measures</i>				
TransferIncidence $_{k,w}$	$\frac{ \mathcal{R}_{k,w}^{\text{transfer}} }{ \mathcal{R}_{k,w} }$	Fraction (0–1)	settlement_transfer_incidence	Fraction of receipt-audited route units $r \in \mathcal{R}_{k,w}$ containing an ERC-20 Transfer log for intermediate vehicle k .
ReceiptCount $_{k,w}$	$ \mathcal{R}_{k,w} $	Route-unit count	settlement_receipt_count	Number of receipt-audited matched route units for vehicle k in UTC week w .